Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (ODD)**	**Course Name**	**Web Technology Lab**
Semester	**3**	**Course Code**	**23CT1301**
Roll No	**13**	**Name of Student**	**Juhi Parwani**

Practical Number	1(A)
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	To develop a To-Do List application using React and TypeScript that allows users to add tasks, mark tasks as completed, delete tasks, and view task statistics.
Problem Definition	Develop a responsive To-Do List web application using React and TypeScript. The application should allow users to enter new tasks, display the list of tasks, mark tasks as completed using checkboxes, and delete tasks when they are no longer required. The application should also display the total number of tasks, completed tasks, and pending tasks, along with a progress bar indicating the completion percentage.
Theory (100 words)	React is a JavaScript library used to build interactive user interfaces through reusable components. TypeScript is a superset of JavaScript that provides static typing and helps identify errors during development. In this practical, React and TypeScript are used to create a To-Do List application. The `Task` interface defines the structure of each task with an ID, title, and completion status. The `useState` Hook manages the list of tasks and generates unique task IDs. Components such as `TaskInput` and `TaskItem` are created to handle task entry and display.

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Procedure and Execution

(100 Words)

Step for Implementation:

Create a React application using Vite with the TypeScript template.
Install the required dependencies using npm.
Create a `types.ts` file and define the `Task` interface.
Create the `TaskInput.tsx` component to accept new tasks.
Create the `TaskItem.tsx` component to display, complete, and delete tasks.
Develop the `App.tsx` component to manage the task list using the `useState` Hook.
Create `App.css` and `index.css` to style the application.
Configure `main.tsx` to render the React application.
Run the application using the Vite development server.
Test adding, completing, and deleting tasks in the browser.

Code:

types.ts

```
export interface Task {
  id: number;
  title: string;
  completed: boolean;
}
```

src/components/TaskInput.tsx

```
iimport { useState, type FormEvent } from "react";

interface TaskInputProps {
  onAddTask: (title: string) => void;
}

function TaskInput({ onAddTask }: TaskInputProps) {
  const [title, setTitle] = useState("");

  const handleSubmit = (event:
FormEvent<HTMLFormElement>) => {
    event.preventDefault();

    const trimmedTitle = title.trim();
```

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```
    if (trimmedTitle) {
      onAddTask(trimmedTitle);
      setTitle("");
    }
  };

  return (
    <form className="task-form" onSubmit={handleSubmit}>
      <input
        type="text"
        placeholder="Enter a new task..."
        value={title}
        onChange={(event) =>
setTitle(event.target.value)}
        aria-label="New task"
        required
      />
      <button type="submit">Add Task</button>
    </form>
  );
}

export default TaskInput;
```

src/components/TaskItem.tsx

```
import type { Task } from "../types";

interface TaskItemProps {
  task: Task;
  onToggle: (id: number) => void;
  onDelete: (id: number) => void;
}

function TaskItem({ task, onToggle, onDelete }:
TaskItemProps) {
```

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```
  return (
    <li className={`task-item ${task.completed ?
"completed" : ""}`}>
      <label className="task-label">
        <input
          type="checkbox"
          checked={task.completed}
          onChange={() => onToggle(task.id)}
        />
        <span>{task.title}</span>
      </label>

      <button
        type="button"
        className="delete-button"
        onClick={() => onDelete(task.id)}
        aria-label={`Delete ${task.title}`}
      >
        Delete
      </button>
    </li>
  );
}

export default TaskItem;
```

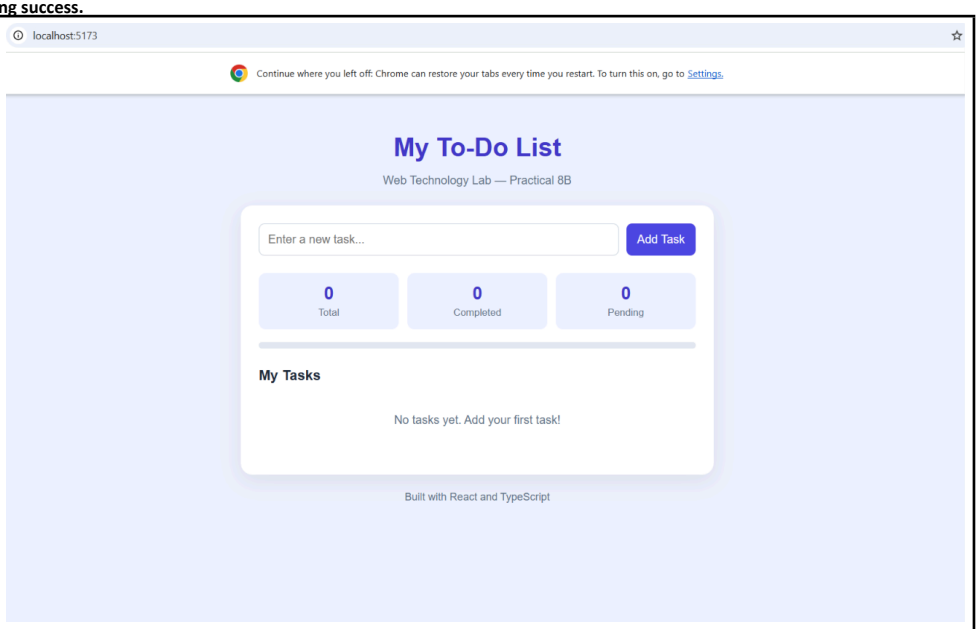
Output:

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Output Analysis	The application successfully performs the required task-management operations. New tasks are added to the list, and checkboxes allow users to mark tasks as completed or pending. The delete button removes unwanted tasks. The total, completed, and pending counts update dynamically whenever the task list changes. The progress bar reflects the percentage of completed tasks. TypeScript interfaces help maintain consistent task data, while React Hooks provide efficient state management and rendering.
Link of student Github profile where lab assignment has been uploaded	**https://github.com/juhiparwa-1507/Web-Technology-Lab.git**
Conclusion	Thus, a To-Do List application was successfully developed using React and TypeScript. The application demonstrates reusable components, TypeScript interfaces, state management with the `useState` Hook, event handling, and dynamic rendering. Users can add, complete, and delete tasks while monitoring task statistics and completion progress.

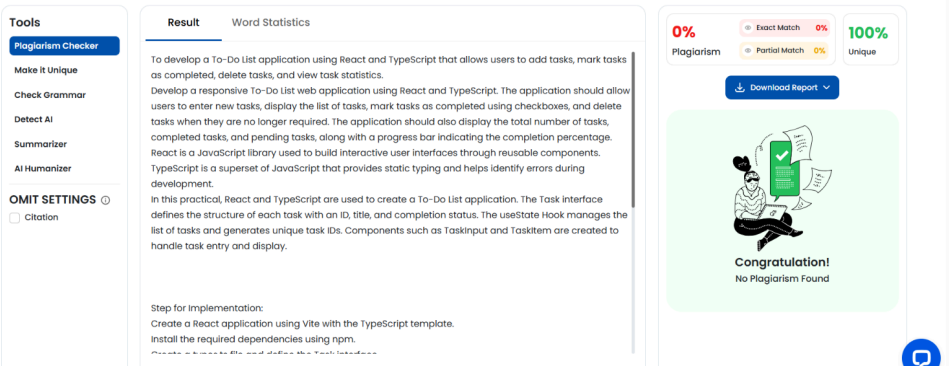
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Date	**14-10-26**